The work-energy Theorem Introduction

The work-energy theorem is best expressed as an equation:

$\Delta K + \Delta U = W_{ext}$

What is W_{ext} ?

- This is the net work done by all external forces
- External means "external to our system"
- $W_{ext} = \Sigma W_i = \Sigma (F_i d_i \cos \theta_i)$ [external forces only]

Examples:

W_{ext} can be:

- work done on an object, or system, by gravity
- work done on an object, or system, by a spring
- work done on an object, or system, by friction
- work done on an object, or system, by some kind of machine (like a person)

Important Notes

Gravitational Force and Energy

- If we define Earth as part of our system, we cannot include the force of gravity as external
- If we define Earth as NOT part of our system, then we cannot have gravitational potential energy
- $|W_{gravity}| = |F_g \Delta y| = |mg \Delta y|$ (we can only determine the sign if we know direction of motion)

Spring Force and Energy

- If we define a spring as part of our system, we cannot include the spring force as external
- If we define a spring as NOT part of our system, then we cannot have spring potential energy

No double-dipping!

Friction and Internal Energy

- For this worksheet, we treat friction as an external force

(Although we will keep in mind that kinetic friction will result in the generation of thermal energy)

- $\Delta U_{int} = f_k d_{relative} \geq 0$ (internal energy is always generated by sliding friction)

The work done by friction on an object depends on the object's displacement. The internal energy generated by friction depends on the relative displacement of the two contacting surfaces.

∴ **Important Considerations**

Is the spring part of the system?

Is Earth part of the system?

Some finer detail

What is a system?

A system is a chosen collection of physical objects. Everything outside it is the surroundings. Forces between objects within the system are internal interactions; forces exerted by external objects are external.

- Block only: forces from Earth, a spring, or a belt are external.
- Block + Earth: gravity is internal; use gravitational potential energy.
- Block + spring: the spring interaction is internal; use elastic potential energy.
- Block + conveyor belt: friction between them is an internal interaction.

What is mechanical energy?

Mechanical energy is the sum of the kinetic and potential energies of a system. It does not include internal (thermal) energy.

$$E_{mech} = K + U \quad \text{(where U may include Ug and Us)}$$

Conservative and nonconservative forces

A conservative force does work that is independent of the path taken between two points. It can be associated with potential energy: $W_{conservative} = -\Delta U$. Examples include gravity and the force of an ideal spring.

A nonconservative force generally does path-dependent work. Sliding friction and air resistance are examples; they can convert mechanical energy into internal energy. Static friction can also transfer mechanical energy between objects without sliding at the contact.

The fuller energy account

For a single object, the work–kinetic-energy theorem is $\Delta K = W_{net}$. A more complete energy balance for a selected system is:

$$\Delta K + \Delta U + \Delta U_{int} = W_{ext}$$

This balance assumes no other energy transfers, such as heat crossing the boundary. In this worksheet, we generally place the contacting frictional surface outside our chosen system and use $\Delta K + \Delta U = W_{ext}$, without separately tracking heating of the objects. This is an idealised mechanical-energy model.

Sliding friction: work and internal-energy generation

Work done by friction on an individual object depends on its displacement: $W_f = f_k d \cos\theta$. It is positive if friction acts with the displacement and negative if it acts against it.

For two surfaces sliding relative to one another, the mechanical energy dissipated at their contact depends instead on the relative sliding distance:

$$\Delta U_{int} = f_k d_{rel} \geq 0$$

In Question 4(c), friction does positive work on the block and negative work on the moving belt. The magnitude of the belt's negative work is greater: the difference becomes internal energy of the contacting objects.

Without slipping, the sliding-friction dissipation is zero. Static friction may still do positive or negative work on an object (Question 4(b)); that does not itself imply thermal energy is generated.

Worked Examples (take g = 10 m/s^2)

1. A block, mass 2.0 kg, is lifted from rest by applying a (vertical) force of 25N over a distance of 60cm

System A diagram: **System B diagram:**

System = block + Earth	System = block
WET: ΔK ΔU W_{ext} ΔME	WET: ΔK ΔU W_{ext} ΔME

2. The person in Q1 now lowers the same block, from rest, applying an upwards force of 15N over 60cm

System = block + Earth	System = block
WET: ΔK ΔU W_{ext} ΔME	WET: ΔK ΔU W_{ext} ΔME

3. A stone is thrown at an angle of 60° to the horizontal at speed of 24 m/s, from horizontal ground level. Neglect air resistance throughout.

a) Determine the horizontal component of the initial velocity

b) Taking the Earth and the stone as the system, determine the relevant terms in the WET (from point of release to the highest point)

c) Hence, assuming no drag forces, determine the maximum height reached by the stone

d) Taking the stone as the system, determine the relevant terms in the WET, and hence determine the maximum height reached by the stone

4. Work done by friction

Example: A box on a moving conveyor belt

(a) Sliding to a stop

Suppose a 5 kg block is moving along a conveyor belt at a speed of 2 m/s, and the belt is suddenly stopped. After initially sliding, the box soon comes to rest. Kinetic friction is 16 N. Assume the belt is fully stationary during the block's subsequent sliding.

Diagram

(i) Taking the block to be the system, determine the relevant terms in the WET, identifying the source(s) of any external work done on the box

(ii) State the internal energy generated as a result of the process

(iii) State the work done by friction on the belt, commenting on what this is likely to be, in reality

(b) Accelerating forwards without sliding

The box, (mass 5 kg) is placed on the initially stationary conveyor belt, which (when switched on) causes the block to accelerate to the right using a constant static friction force of 20 N to the right, over a distance of 0.5 m

Determine:

i) The work done on the block by friction (by the belt)

ii) Hence, the KE gained by the block, and final speed of the block

iii) The internal energy generated by friction at the contact

iv) The work done on the belt by friction (by the block)

(c) Accelerating forwards with sliding

Now suppose that the stationary block is dropped (from a tiny height) on the moving conveyor belt, so that the block is initially accelerated by kinetic friction, of 0.8 N. Suppose that the block moves 10 cm forward, relative to the ground, during this process, at which point the block has matched the speed of the belt. Assume the belt maintains a constant speed throughout the sliding interval.

i) Determine the work done on the block by friction (by the belt)

Hence,

ii) determine the KE gained by the block, and final speed of the block

iii) determine the internal energy generated by sliding friction

iv) the work done on the belt by friction (by the block)

v) Draw an energy diagram to illustrate the flow of energy taking place